CHOOSING THE RIGHT NETWORK TECHNOLOGY

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INTRODUCTION

MORE THAN 4 BILLION PEOPLE WILL USE INTERNET BY END OF 2018 (KEMP, 2018)

USED FOR

- COMMUNICATION
- MARKETING
- IMPROVING PRODUCTIVITY
- PROVIDING CUSTOMER SUPPORT
- CLOUD DATA STORAGE

https://www.howtogeek.com/wp-content/uploads/2018/02/img_5a78dece9a202.jpg

SITUATION CHALLENGE

- ESSENTIALLY A WAN
- CONNECT GEOGRAPHICALLY SEPARATED BUSINESS LOCATIONS
- APPLICATIONS PRODUCE LARGE NETWORK VOLUME TRAFFIC
- NEED FOR MORE BANDWIDTH BECAUSE OF
  - DESKTOP VIRTUALIZATION
  - VOICE OVER INTERNET PROTOCOL (VOIP)
  - VIDEO APPLICATIONS
  - REMOTE WORKERS

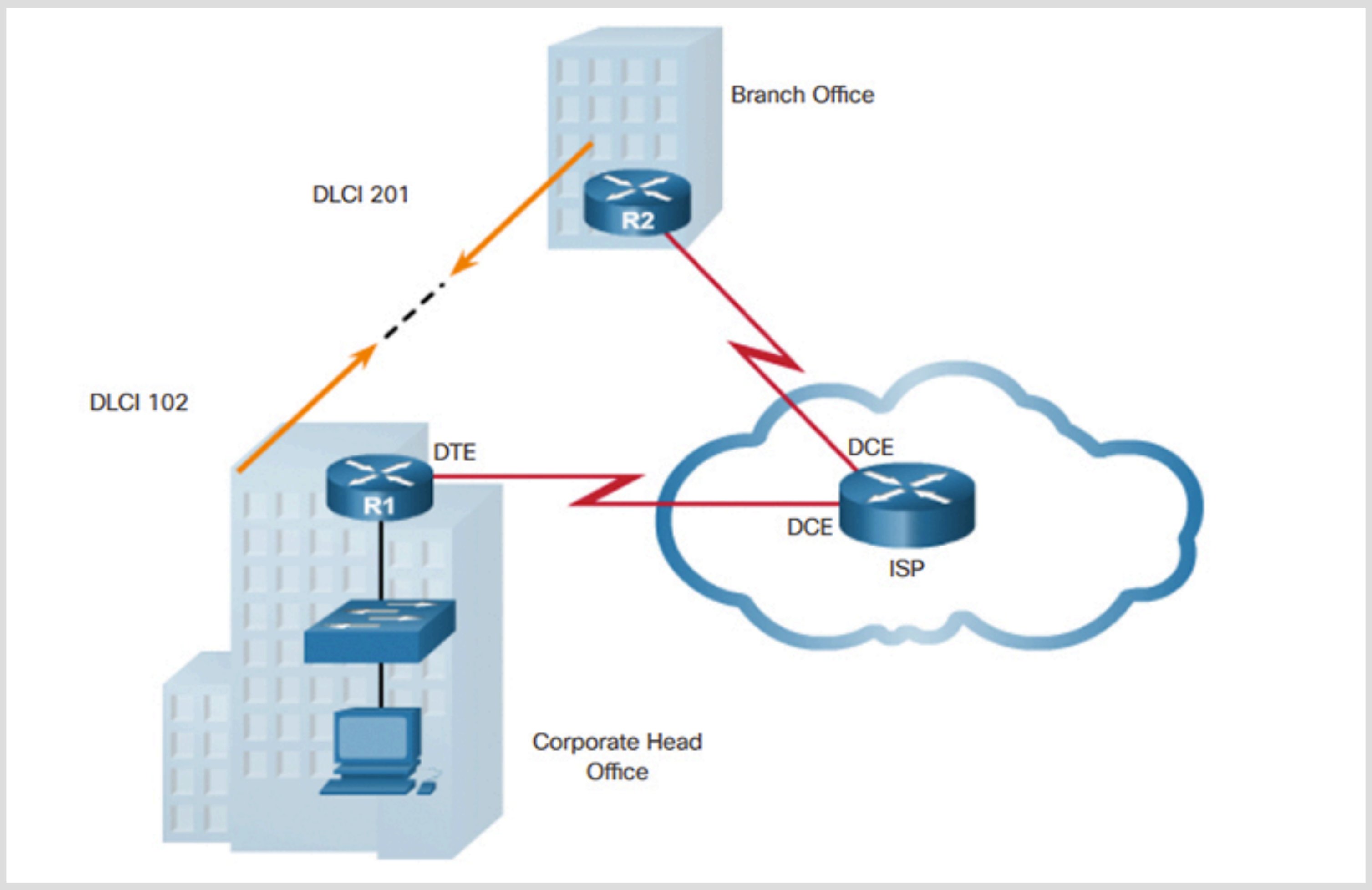
SOLUTION OPTIONS

- HIGHEST SUCCESS WITH IMPLEMENTATION LIES IN UNDERLYING INFRASTRUCTURE
- THOROUGH ANALYSIS OF NETWORK SERVICES NEEDED
- BUSINESSES LOOK FOR SPECIAL TECHNOLOGY / ALTERNATIVES TO WAN SUCH AS
  - FRAME RELAY
  - ASYNCHRONOUS TRANSFER MODE (ATM)
  - MULTIPROTOCOL LABEL SWITCHING (MPLS)
  - CARRIER ETHERNET

FRAME RELAY

- A DATA LINK LAYER, DIGITAL PACKET SWITCHING NETWORK PROTOCOL TECHNOLOGY DESIGNED TO CONNECT LOCAL AREA NETWORKS (LANS) AND TRANSFER DATA ACROSS WIDE AREA NETWORKS" (MITCHELL, 2018).
- SENDS DATA IN VARIABLE-SIZED UNITS CALLED *FRAMES* (PACKET LAB, 2010)
- RELIES ON ENDPOINTS TO PROVIDE ERROR-CORRECTION
  - THIS SPEEDS TRANSMISSION BUT LEAVES THE ENDPOINTS RESPONSIBLE FOR DETECTING AND RETRANSMITTING DROPPED FRAMES (PACKET LAB, 2010)
- COST-EFFECTIVE METHOD IT IS NO LONGER CONSIDERED ONE OF THE TOP METHODS

SAMPLE FRAME RELAY TOPOLOGY



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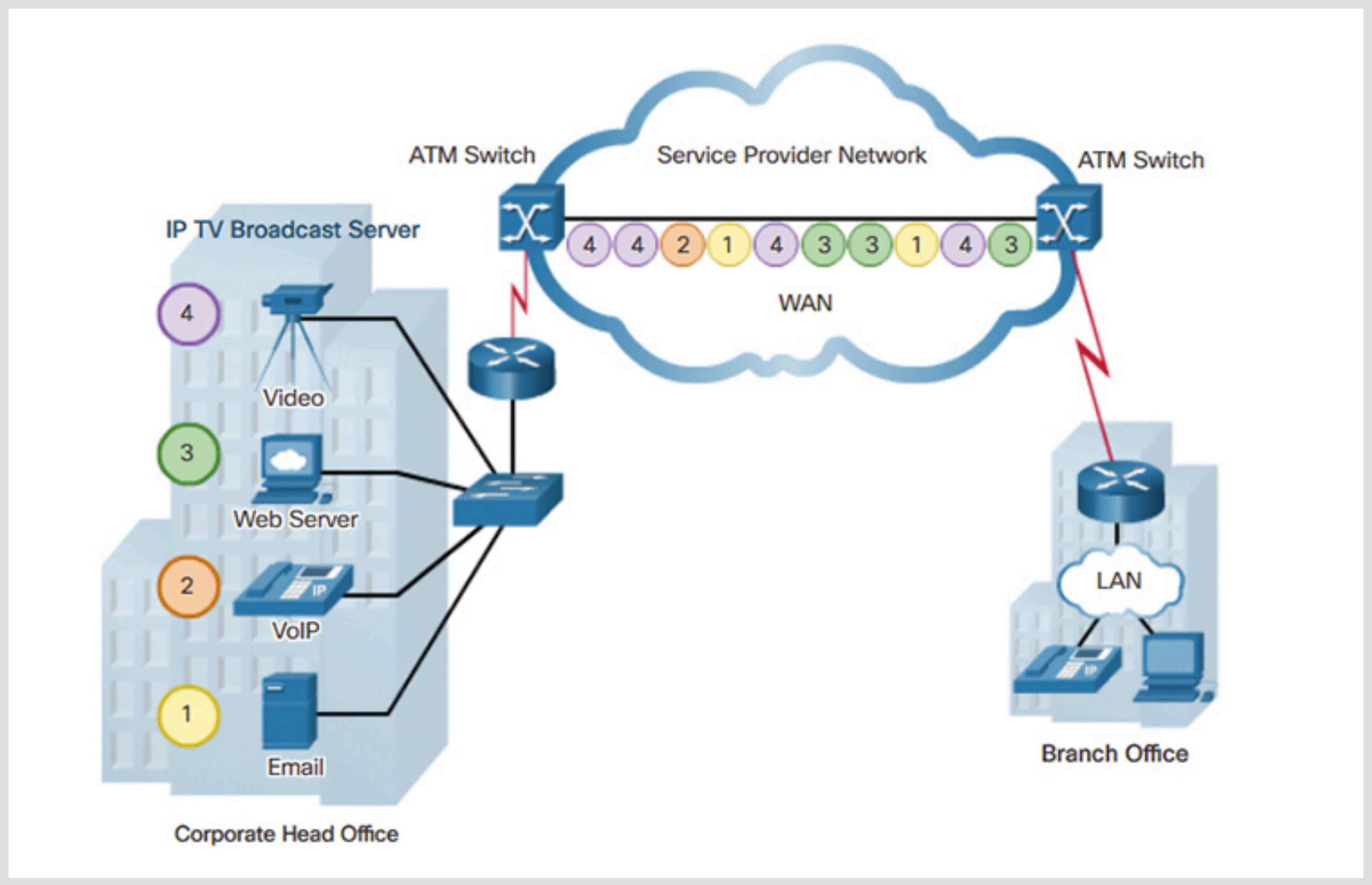
ATM

- “A TELECOMMUNICATION CONCEPT, DEVELOPED FOR THE DATA LINK LAYER, TO CARRY ALL SORTS OF DIFFERENT KINDS OF DATA SUCH AS WEB TRAFFIC, VOICE, VIDEO, ETC. WHILE PROVIDING QOS (QUALITY OF SERVICE) AT THE SAME TIME,” (SHEKHAR, 2017)
- ACHIEVES THE ABILITY TO TRANSFER THIS VARIETY OF TRAFFIC TYPES THROUGH ITS USE OF LOGICAL CONNECTIONS THROUGH VIRTUAL PATHS AND VIRTUAL CHANNELS (STALLINGS & CASE, 2013, P. 467)

SAMPLE ATM TOPOLOGY



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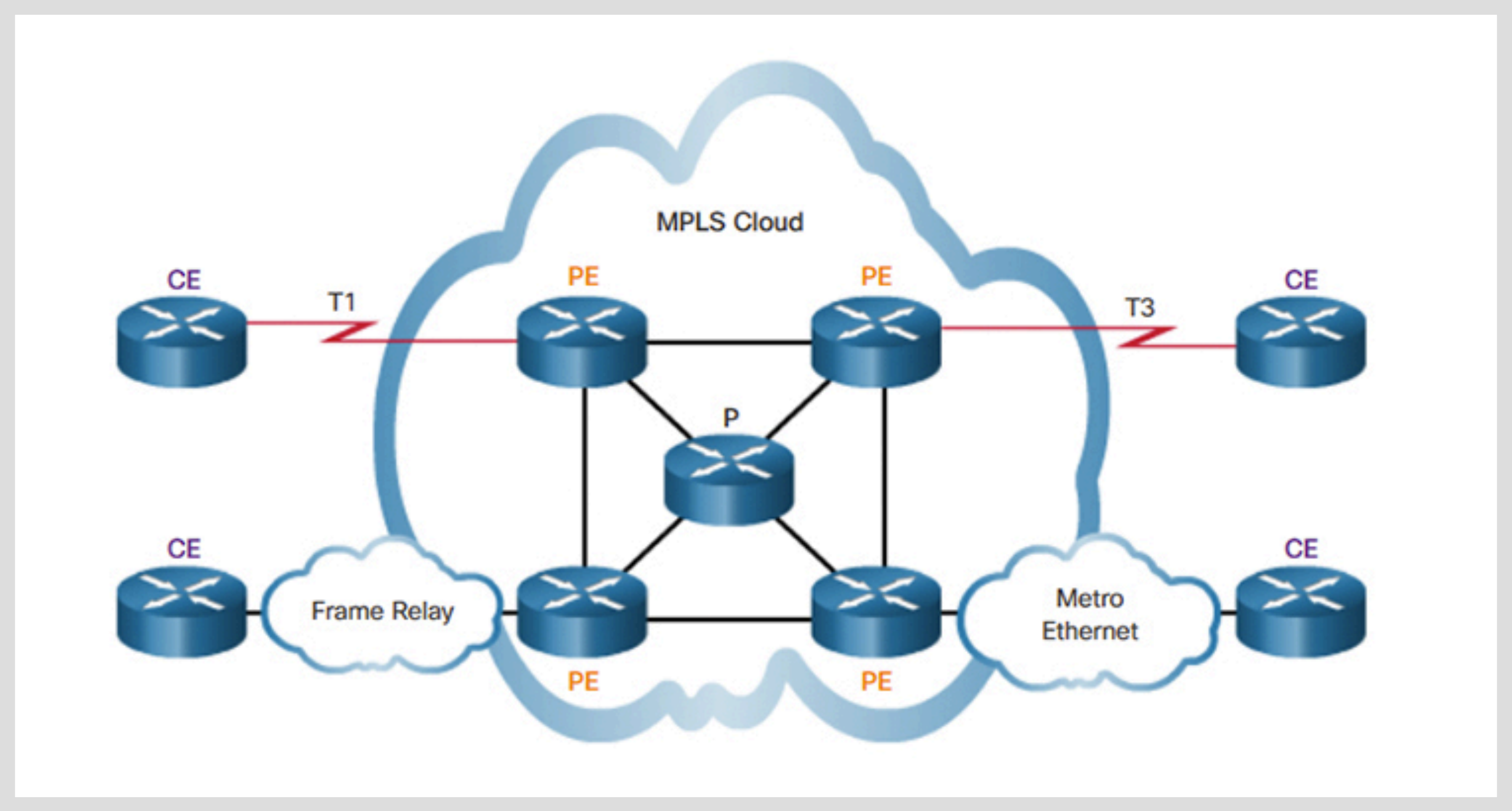
MULTIPROTOCOL LABEL SWITCHING (MPLS)

- “A WAY TO INSURE RELIABLE CONNECTIONS FOR REAL-TIME APPLICATIONS” (WEINBERG & JOHNSON, 2018, PARA. 1)
- COMMONLY USED WITH OFFICES, BIG NETWORKS, AND ANY PLACE THAT NEEDS QUALITY OF SERVICE WITH REAL-TIME APPLICATIONS
- “MUST BE PURCHASED FROM A CARRIER AND IS FAR MORE EXPENSIVE THAN SENDING TRAFFIC OVER THE PUBLIC INTERNET” (WEINBERG & JOHNSON, 2018, PARA. 23)

SAMPLE MPLS TOPOLOGY



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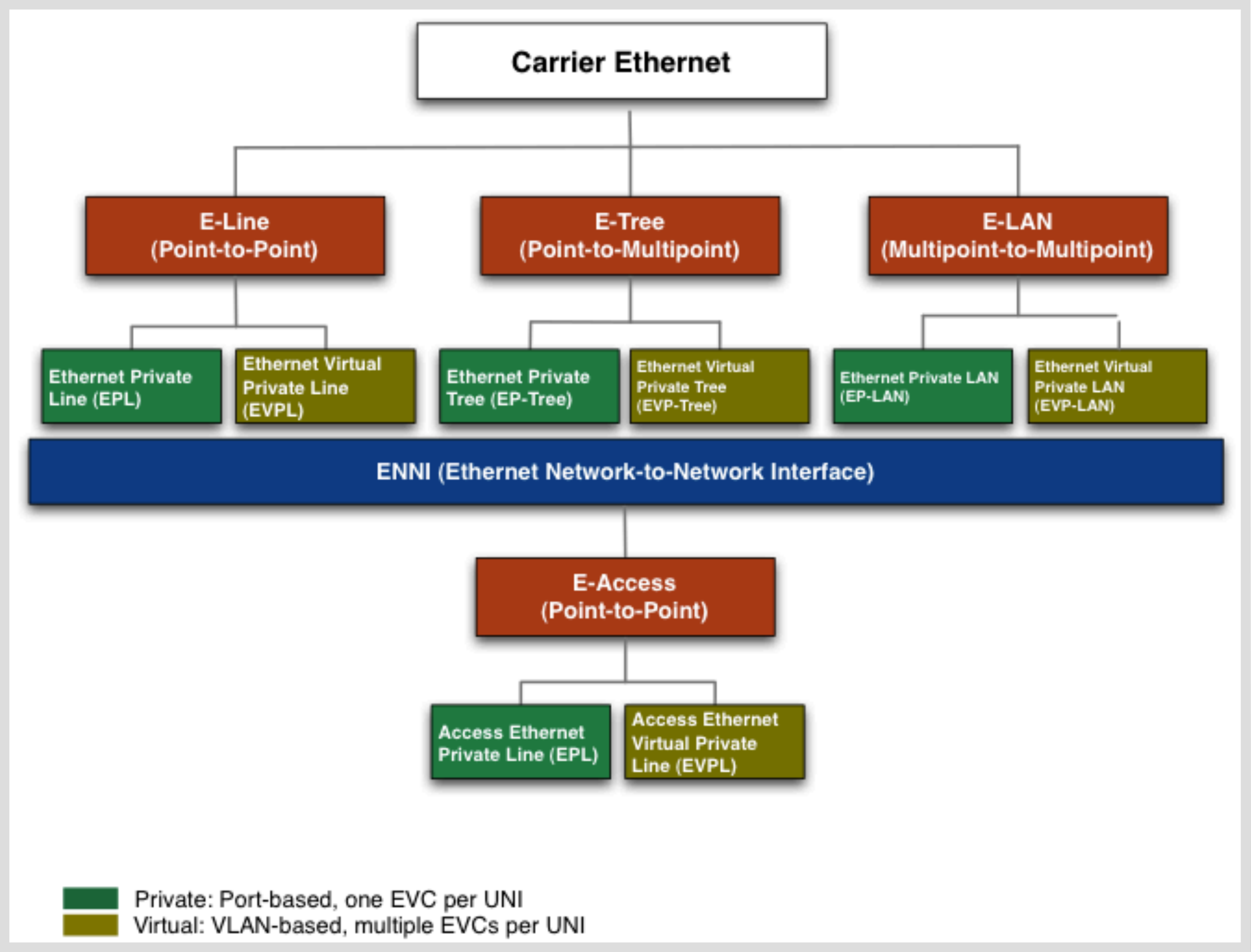
CARRIER ETHERNET

- “ENABLES A VARIETY OF NETWORK SERVICES THAT CAN BE SOLD TO BOTH OTHER CARRIERS (WHOLESALE) AND END USERS (RETAIL)” (HAWKINS, 2016)
- “A FEATURE-RICH SOLUTION PERFECT FOR ALL NETWORK OPERATORS (CARRIERS, ENTERPRISES, AND GOVERNMENT) WHO PROVIDE NETWORKING SERVICES TO END USERS, BOTH INTERNALLY AND EXTERNALLY” (HAWKINS, 2016)
- CARRIER ETHERNET SERVICES ARE LOWER COST SOLUTIONS BUT ARE TYPICALLY FASTER THAN ATM AND FRAME RELAY

CARRIER ETHERNET TOPOLOGY



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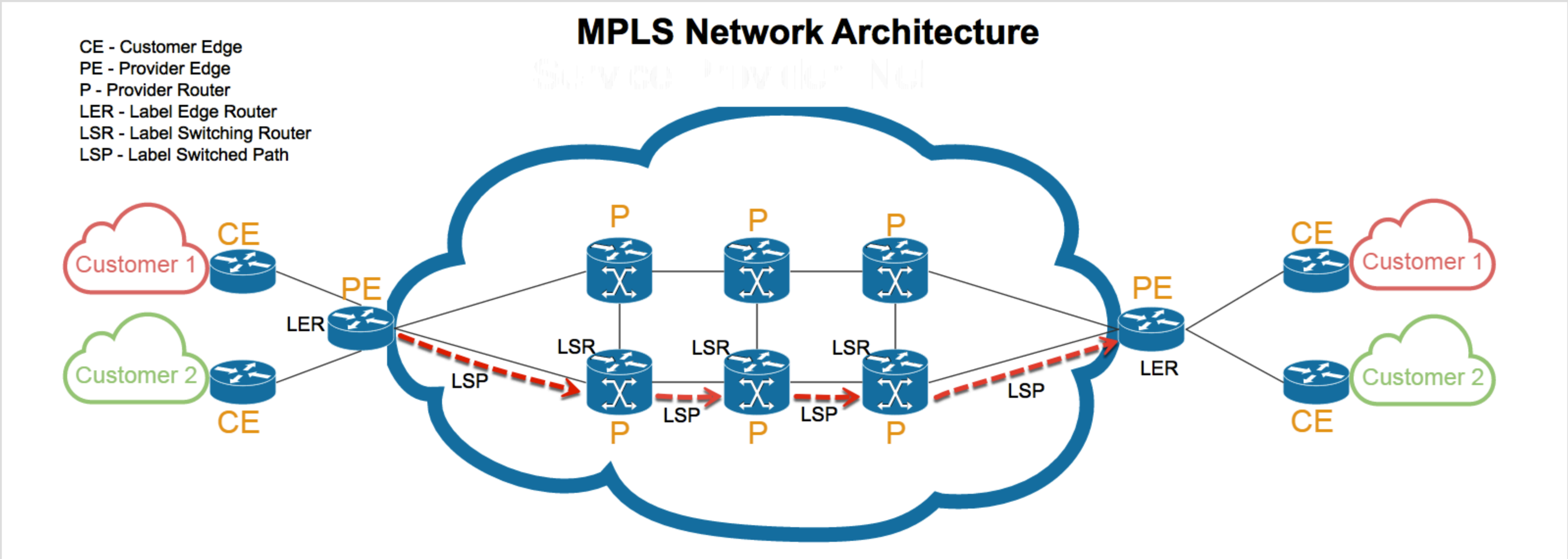
SCENARIO A

- LARGE NUMBER OF LOCATIONS IN A METROPOLITAN AREA
- AT EACH LOCATION, REAL-TIME DATA TRANSACTIONS ARE EXPECTED TO BE SENT AND PROCESSED INDEPENDENTLY AND RANDOMLY
- TO ACHIEVE REAL-TIME PERFORMANCE, DELAY MUST BE KEPT TO A MINIMUM
- THE DATA VOLUMES ARE EXPECTED TO BE MODEST, BUT TRANSFER RATES MAY REACH SEVERAL MEGABITS PER SECOND (MBPS)

SCENARIO A SUGGESTION

- WE SUGGEST USING MPLS FOR REAL-TIME APPLICATIONS
- MORE EXPENSIVE THAN OTHER OPTIONS, BUT HAS THE MOST RELIABILITY FOR REAL-TIME PERFORMANCE
- SINCE EACH LOCATION HAS MANY REAL-TIME TRANSACTIONS TO PROCESS, THIS COMPONENT IS VERY IMPORTANT



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SCENARIO B

- WAN THAT INTERCONNECTS SIX OR MORE LOCATIONS IN REMOTE AREAS ACROSS THE UNITED STATES
- TRANSMISSION FACILITIES ARE VARIED BUT INCLUDE:
  - RADIO LINKS
  - SATELLITE LINKS
  - PHONE LINKS USING MODEMS

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SCENARIO B SUGGESTION

- WITH MODEST DATA RATES SUPPORTED BY THESE TRANSMISSION LINKS, ATM WOULD BE A GOOD CHOICE
- SMALL, FIXED SIZE CELLS USED IN ATM COMMUNICATION OFFER SEVERAL ADVANTAGES, INCLUDING:
  - DATA PRIORITIZATION
  - EFFICIENT CELL SWITCHING
  - SIMPLE IMPLEMENTATION OF THE SWITCHING MECHANISM IN HARDWARE (STALLINGS AND CASE, 2013)
- SUPPORTS REAL-TIME AND NON-REAL-TIME SERVICE AT CONSTANT OR VARIABLE FRAME RATES
- GUARANTEED FRAME RATE (GFR) SERVICE CATEGORY ENABLES THE OPTIMIZATION OF TRAFFIC PASSING FROM A LAN ROUTER ONTO A BACKBONE NETWORK (STALLINGS & CASE, 2013)

SCENARIO C

- MULTIMEDIA APPLICATION NETWORK SERVICING A SMALL NUMBER OF LOCATIONS
  - INCLUDES IMAGE SHARING AND REAL-TIME AUDIO AND VIDEO SERVICES
  - OTHER DATA SERVICES MAY BE INTERSPERSED AS WELL
- HIGH VOLUMES OF TRAFFIC ARE EXPECTED TO REQUIRE TRANSFER RATES IN GIGABITS PER SECOND (GBPS) RANGES
- ONLY A FEW LOCATIONS WILL BE CONNECTED, BUT MANY VIRTUAL CIRCUITS WILL BE NEEDED TO ACCOMMODATE MANY USERS AND LARGE DATA SIZES

SCENARIO C SUGGESTION

- INTERACTIVE MULTIMEDIA APPLICATIONS HAVE STRINGENT REQUIREMENTS FOR LOSS AND DELAY
- STREAM-BASED APPLICATIONS DELIVER DATA AT A VARIABLE RATE, PARTICULARLY WHEN CONSTANT QUALITY IS REQUIRED
- ATM CAN INTEGRATE TRAFFIC OF DIFFERENT NATURES WITHIN A SINGLE NETWORK CREATING INTERACTIONS OF DIFFERENT TYPES THAT REDUCE DELAY JITTER AND LOSS
- TRADITIONAL PROTOCOL LAYERS DO NOT HAVE THE APPROPRIATE MECHANISMS TO PROVIDE THE REQUIRED NETWORK QUALITY OF SERVICE (QOS) FOR SUCH INTERACTIVE VARIABLE BIT RATE (VBR) MULTIMEDIA MULTIPOINT APPLICATIONS (ADANEZ, FRANCISCO, & HUBAUX, 1998)

CONCLUSION

- INVESTIGATING AVAILABLE SERVICES WE ARE ABLE TO RECOMMEND A SUITABLE SOLUTION
- APPROACH CAN BE APPLIED TO ANY NETWORK DESIGN SCENARIO:
  - IDENTIFY REQUIREMENTS
  - RESEARCH AVAILABLE TECHNOLOGIES
  - DOCUMENT RESULTS
  - PROVIDE DETAILED RECOMMENDATION

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